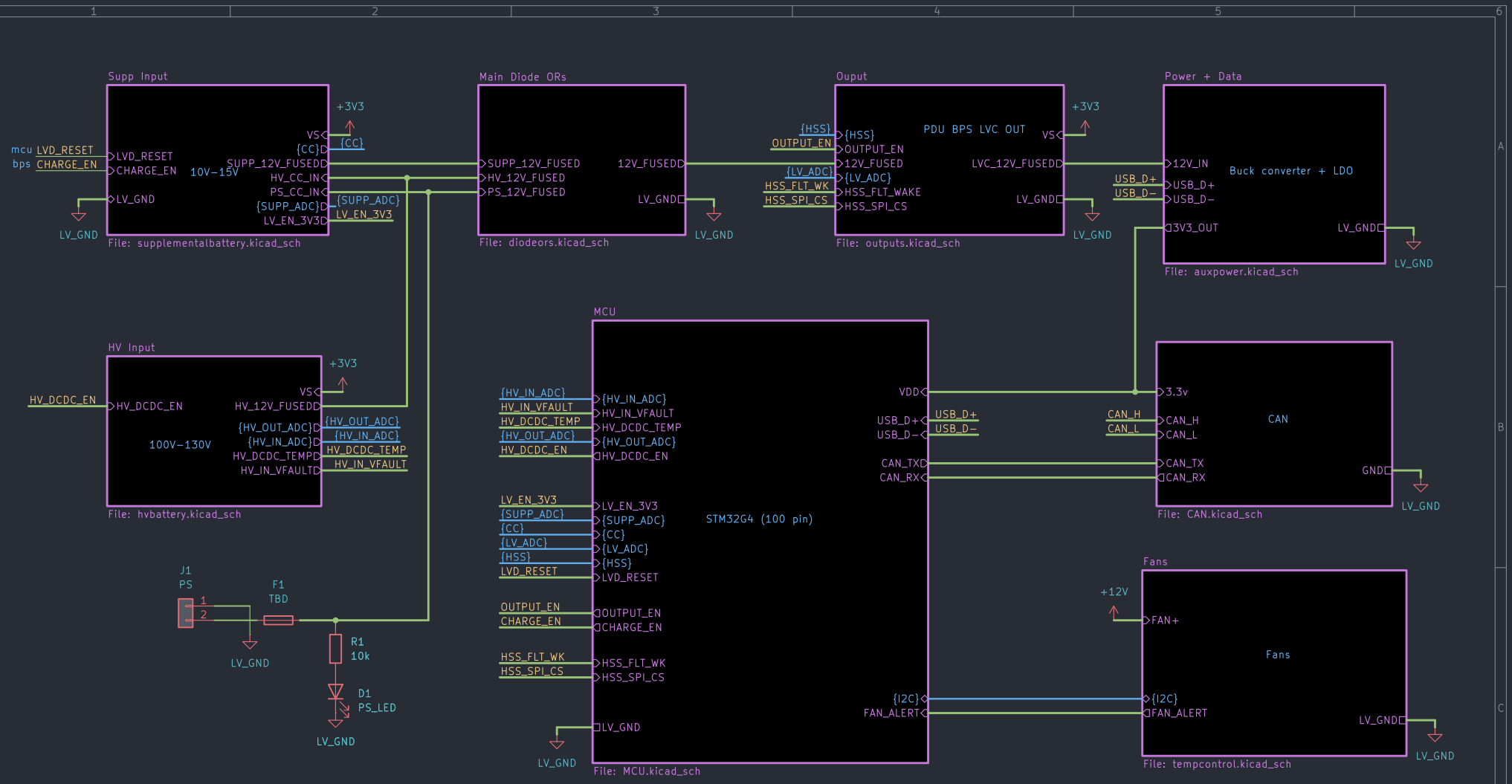


Next Revision Notes

update ideal diode controller TVS to standard bidirectional
DNP shield GND 0 ohm. desolder before testing

H1 MountingHole
H2 MountingHole
H3 MountingHole

CAN_H 1
CAN_L 2
SHIELD 3
CAN_H
CAN_L
SHIELD
J11
CAN_IN

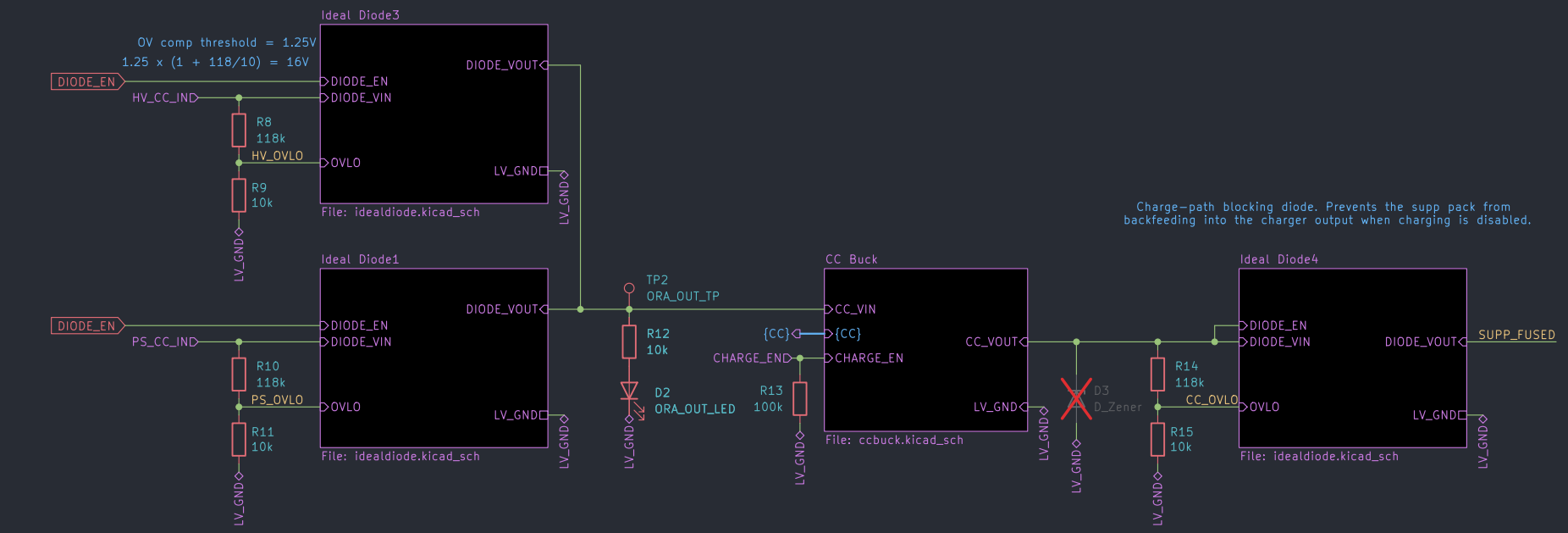
SHIELD
R5
0
LV_GND

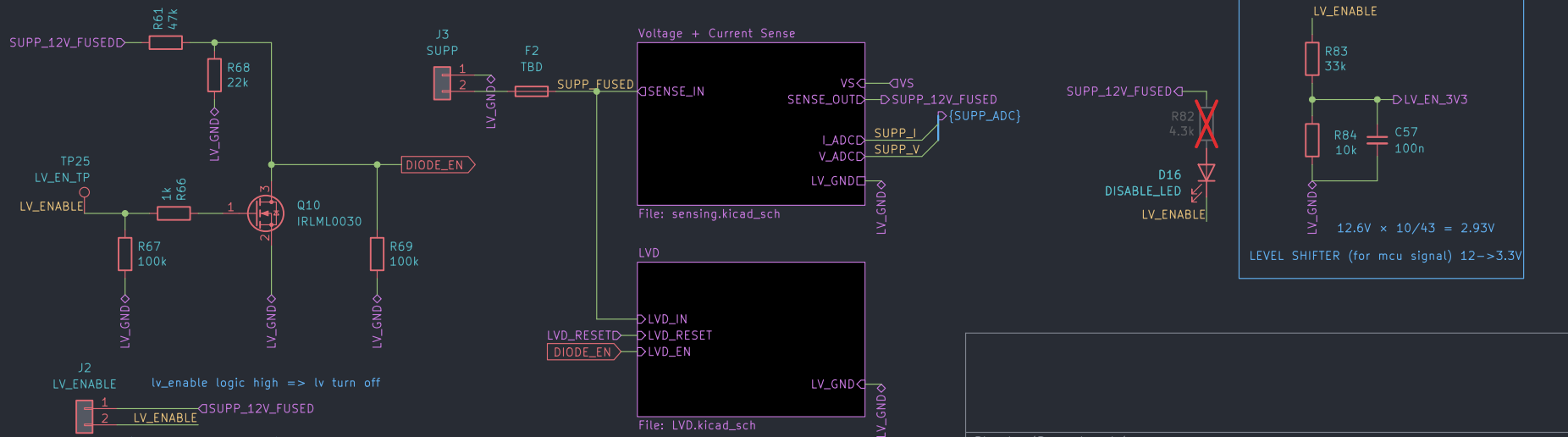
CAN_H 1
CAN_L 2
SHIELD 3
CAN_H
CAN_L
SHIELD
J18
CAN_OUT

Jennifer Li LHRS
Sheet: /
File: PS–LVCarrier.kicad_sch

Title: LV Carrier

Size: A4	Date: 7/2026	Rev: A
KiCad E.D.A. 10.0.3		Id: 1/29

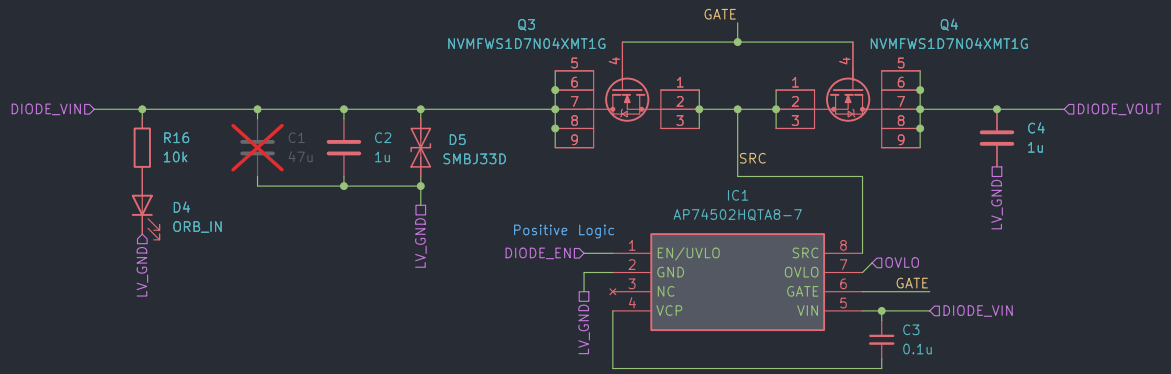




Sheet: /Supp Input/
File: supplementalbattery.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/29



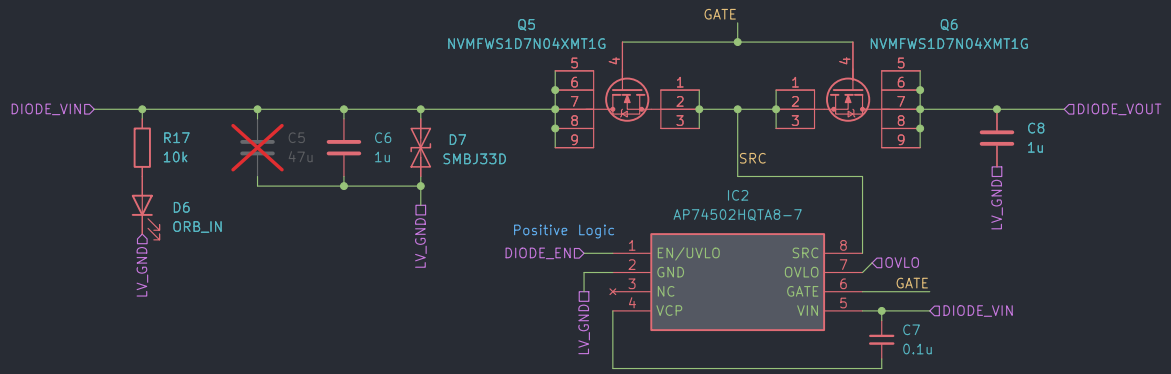
TP3
DIODE_VIN

DIODE_VIN

Sheet: /Supp Input/Ideal Diode3/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 3/29

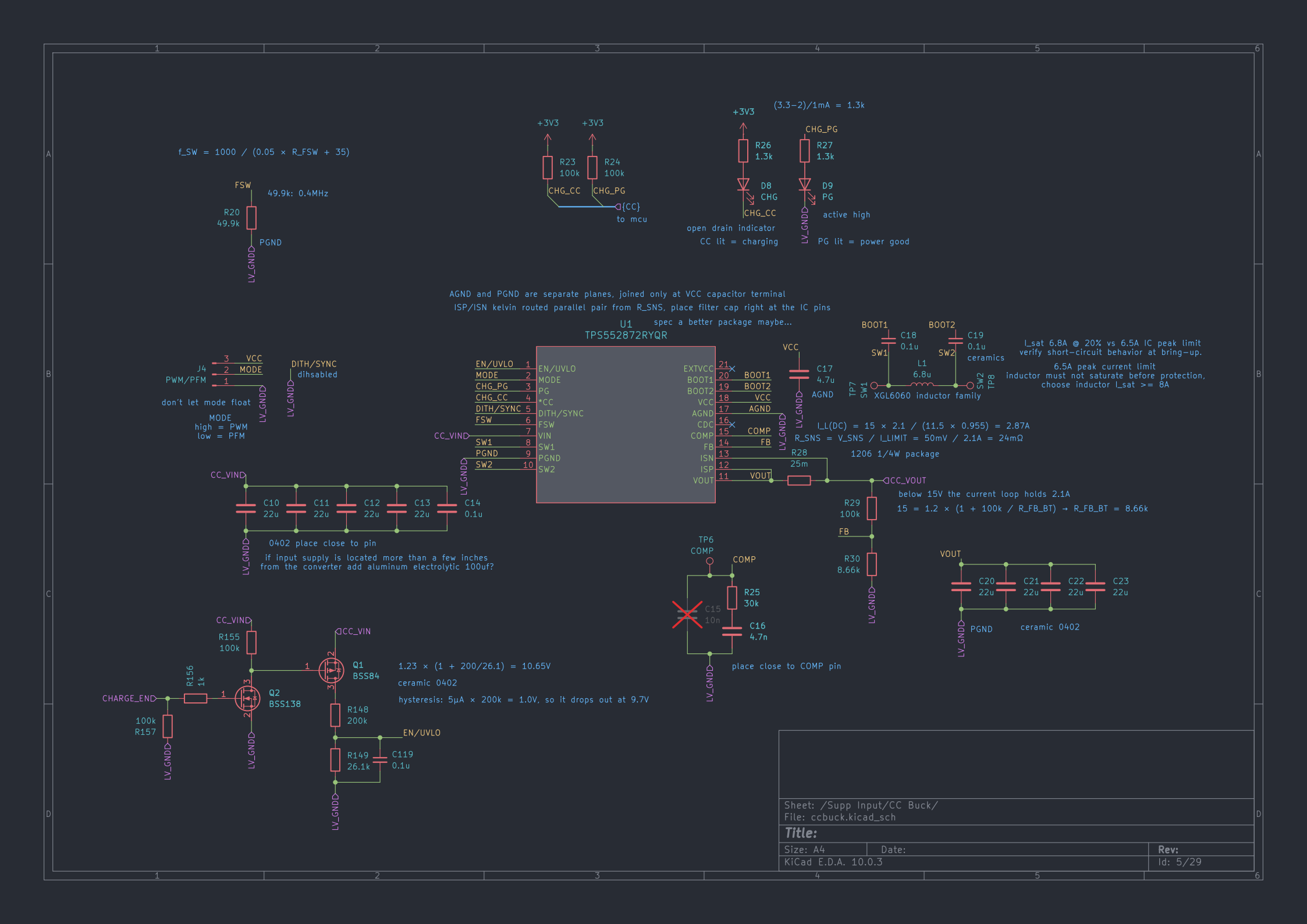


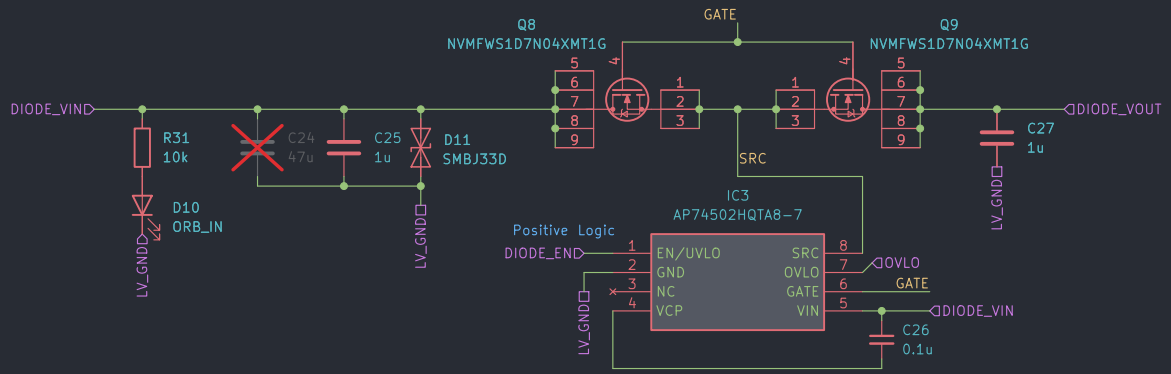
TP4
DIODE_VIN
DIODE_VIN

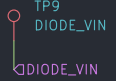
Sheet: /Supp Input/Ideal Diode1/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 4/29


f_SW = 1000 / (0.05 × R_FSW + 35)
FSW
49.9k: 0.4MHz
R20
49.9k
PGND
LV_GNDD
+3V3
+3V3
R23
100k
R24
100k
CHG_CC
CHG_PG
{CC}
to mcu
(3.3−2)/1mA = 1.3k
+3V3
R26
1.3k
CHG_PG
R27
1.3k
D8
CHG
D9
PG
CHG_CC
active high
open drain indicator
CC lit = charging
PG lit = power good
AGND and PGND are separate planes, joined only at VCC capacitor terminal
ISP/ISN kelvin routed parallel pair from R_SNS, place filter cap right at the IC pins
U1
spec a better package maybe...
TPS552872RYQR
EN/UVLO 1
MODE 2
CHG_PG 3
CHG_CC 4
DITH/SYNC 5
FSW 6
7
SW1 8
PGND 9
SW2 10
EXTVCC 21
BOOT1 20
BOOT2 19
VCC 18
AGND 17
16
COMP 15
FB 14
13
ISN 12
ISP 12
VOUT 11
J4
3 VCC
2 MODE
1
PWM/PFM
don't let mode float
MODE
high = PWM
low = PFM
DITH/SYNC
dihsabled
CC_VIND
C10
22u
C11
22u
C12
22u
C13
22u
C14
0.1u
0402 place close to pin
if input supply is located more than a few inches
from the converter add aluminum electrolytic 100uf?
VCC
C17
4.7u
AGND
BOOT1
C18
0.1u
BOOT2
C19
0.1u
ceramics
SW1
SW2
L1
6.8u
XGL6060 inductor family
TP7
TP8
I_sat 6.8A @ 20% vs 6.5A IC peak limit
verify short−circuit behavior at bring−up.
6.5A peak current limit
inductor must not saturate before protection,
choose inductor I_sat >= 8A
I_L(DC) = 15 × 2.1 / (11.5 × 0.955) = 2.87A
R_SNS = V_SNS / I_LIMIT = 50mV / 2.1A = 24mΩ
1206 1/4W package
R28
25m
CC_VOUT
below 15V the current loop holds 2.1A
15 = 1.2 × (1 + 100k / R_FB_BT) → R_FB_BT = 8.66k
R29
100k
FB
R30
8.66k
TP6
COMP
R25
30k
C15
10n
C16
4.7n
place close to COMP pin
VOUT
C20
22u
C21
22u
C22
22u
C23
22u
PGND
ceramic 0402
CC_VIND
R155
100k
R156
1k
CHARGE_END
100k
R157
Q1
BSS84
Q2
BSS138
CC_VIN
1.23 × (1 + 200/26.1) = 10.65V
ceramic 0402
hysteresis: 5µA × 200k = 1.0V, so it drops out at 9.7V
R148
200k
EN/UVLO
R149
26.1k
C119
0.1u
Sheet: /Supp Input/CC Buck/
File: ccbuck.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 5/29

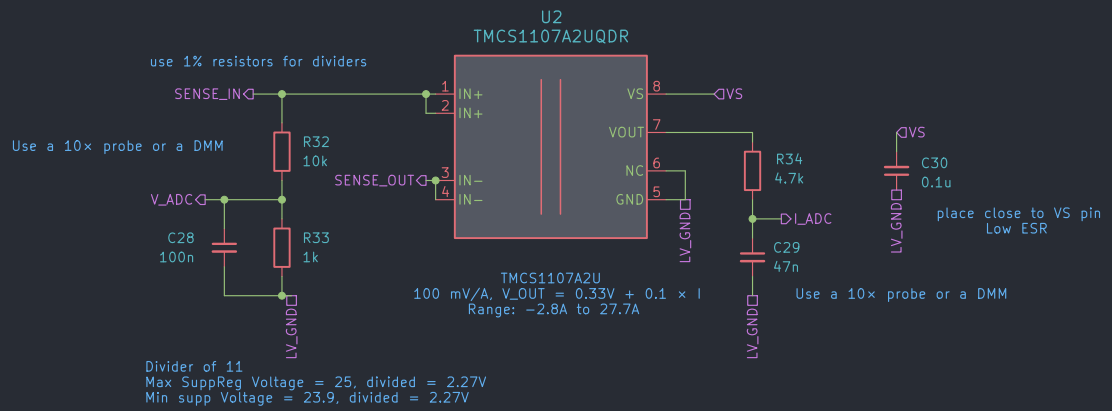




Sheet: /Supp Input/Ideal Diode4/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/29



Sheet: /Supp Input/Voltage + Current Sense/
File: sensing.kicad_sch

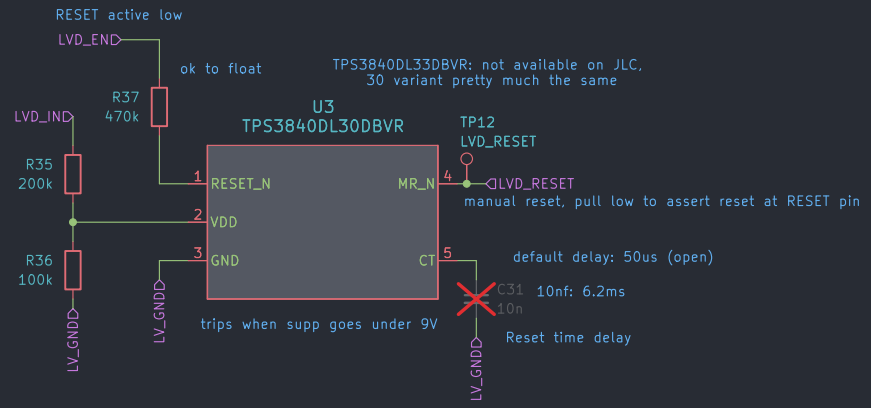
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/29

6.6 Timing Requirements

At 1.5V ≤ V_{DD} ≤ 10V, CT = $\overline{MR}$ = Open, $\overline{RESET}$ pull-up resistor ($R_{pull-up}$) = 100kΩ to VDD, output reset load (C_{LOAD}) = 10pF and over the operating free-air temperature range – 40°C to 125°C, VDD slew rate < 100mV / us, unless otherwise noted. Typical values are at T_J = 25°C.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{STRT}	Startup Delay(1)	CT pin open	100	220	350	µs
t_{P_HL}	Propagation detect delay for VDD falling below V_{IT-}	V_{DD} = V_{IT+} to (V_{IT-}) - 10%(2)		15	30	µs
t_D	Reset time delay(3)	CT pin = open			50	µs
		CT pin = 10nF		6.2		ms
		CT pin = 1 µF		619		ms
t_{GI_VIT-}	Glitch immunity V_{IT-}	5% V_{IT-} overdrive(4)		10		µs
t_{MR_PW}	$\overline{MR}$ pin pulse duration to initiate reset			300		ns
t_{MR_RES}	Propagation delay from $\overline{MR}$ low to reset	V_{DD} = 4.5V, $\overline{MR}$ < $V_{\overline{MR}_L}$		700		ns
t_{MR_tD}	Delay from release $\overline{MR}$ to deasert reset	V_{DD} = 4.5V, $\overline{MR}$ = $V_{\overline{MR}_L}$ to $V_{\overline{MR}_H}$		t_D		ms



Vit− = 3V

3.0 × (200k + 100k) / 100k = 3.0 × 3.0 = 9.00V

V_rail_max = 10 × (200k + 100k) / 100k = 30V

8.2.3.2 Detailed Design Procedure

The primary constraint for this application is monitoring a 12V rail while preventing the VDD pin on TPS3840 from exceeding the recommended maximum of 10V. This is accomplished by sizing the resistor divider so that when the 12V rail drops to 10V, the VDD pin for TPS3840 is at 4.9V which is the V_{IT-} threshold for triggering a undervoltage condition for TPS3840DL49 as shown in Equation 8.

$$V_{rail_trigger} = V_{IT-} \times \left(R_{top} + R_{bottom}\right) \div R_{bottom} \quad (8)$$

where $V_{rail_trigger}$ is the trigger voltage of the rail being monitored, V_{IT-} is the falling threshold on the VDD pin of TPS3840, and R_{top} and R_{bottom} are the top and bottom resistors of the external resistor divider. Be sure to size the resistor values such that the current through the external resistor divider is much greater than IDD to preserve voltage monitoring accuracy. V_{IT-} is fixed per device variant and is 4.9V for TPS3840DL49. Substituting in the values from Figure 8-5, the undervoltage trigger threshold for the rail is set to 10.045V.

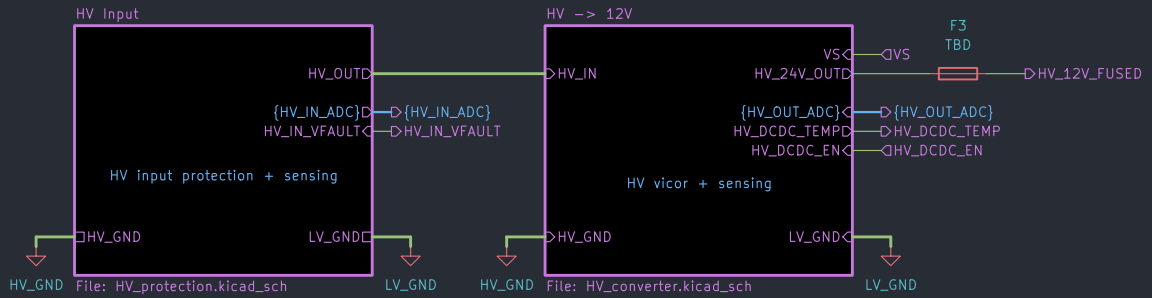
Since the undervoltage trigger of 10V on the rail corresponds to 4.9V undervoltage threshold trigger of the TPS3840 device, there is plenty of room for the rail to rise up while maintaining less than 10V on the VDD pin of the TPS3840. Equation 9 shows the maximum rail voltage that still meets the 10V maximum at the VDD pin for TPS3840.

$$V_{rail_max} = 10 \times (10500 + 10000) \div 10000 = 20.5V \quad (9)$$

Sheet: /Supp Input/LVD/
File: LVD.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 8/29



Sheet: /HV Input/		
File: hvbattery.kicad_sch		
Title:		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 9/29

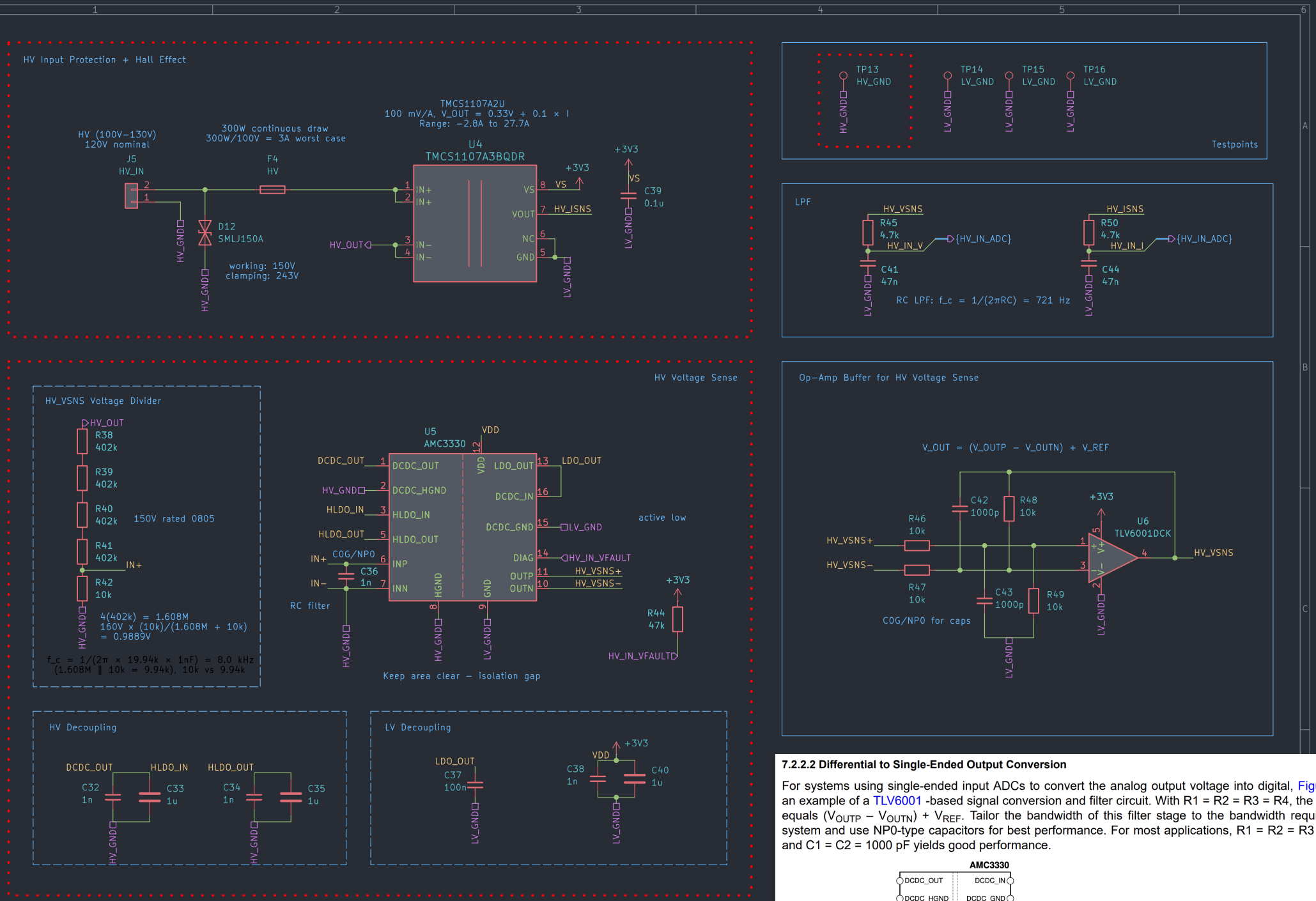
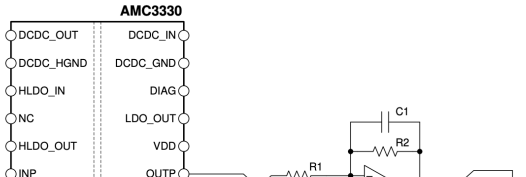


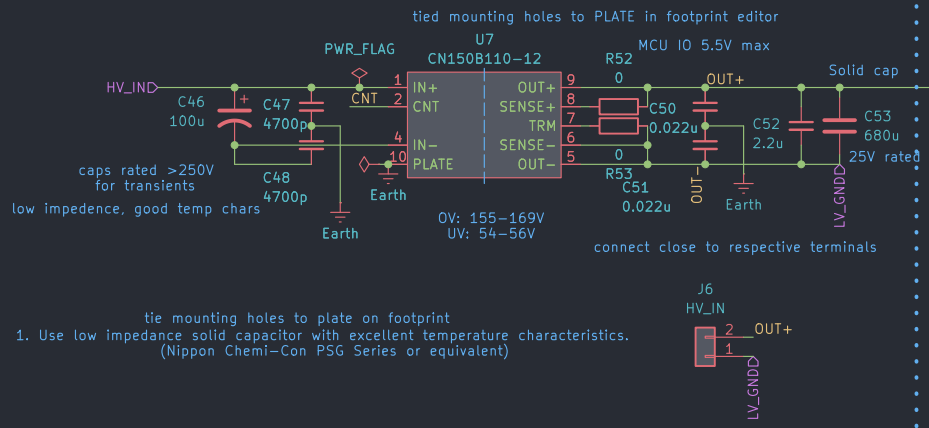
Table 7-1. Design Requirements

7.2.2.2 Differential to Single-Ended Output Conversion

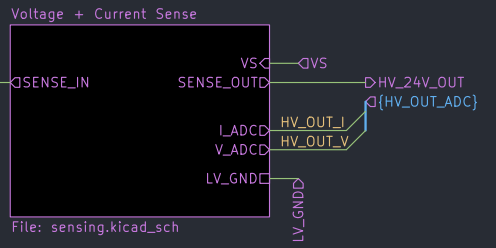
For systems using single-ended input ADCs to convert the analog output voltage into digital, Figure 7-3 an example of a TLV6001 -based signal conversion and filter circuit. With R1 = R2 = R3 = R4, the output v equals (V_{OUTP} – V_{OUTN}) + V_{REF}. Tailor the bandwidth of this filter stage to the bandwidth requirement system and use NP0-type capacitors for best performance. For most applications, R1 = R2 = R3 = R4 = and C1 = C2 = 1000 pF yields good performance.



High Voltage DCDC

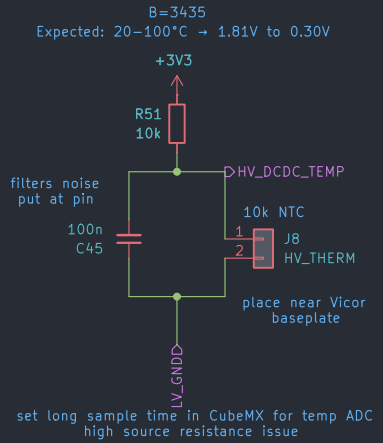


tie mounting holes to plate on footprint
1. Use low impedance solid capacitor with excellent temperature characteristics.
(Nippon Chemi-Con PSG Series or equivalent)



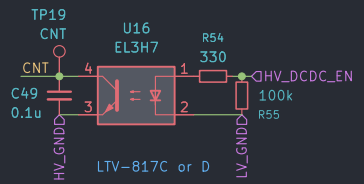
High Voltage 24V Ouput VIP Sensing

HV Vicor Temperature Sensing



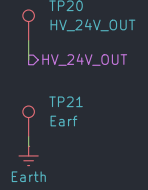
set long sample time in CubeMX for temp ADC
high source resistance issue

Primary Control (HV isolated) opto disable



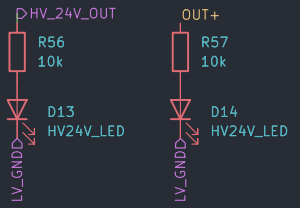
led off -> transistor off,
CNT not pulled to -VIN,
internal pullup pulls CNT
high -> converter off

Testpoints



Indicator LEDs

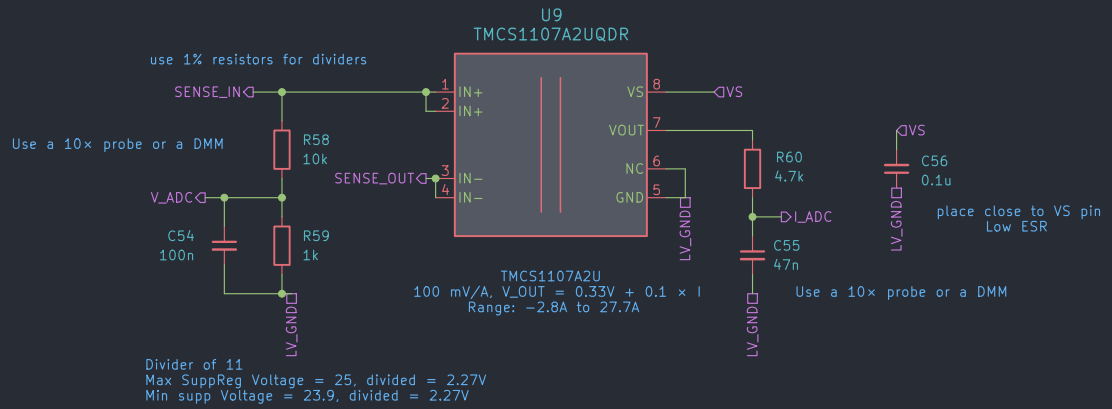
add indicator led math to all



Sheet: /HV Input/HV -> 12V/
File: HV_converter.kicad_sch

Title:

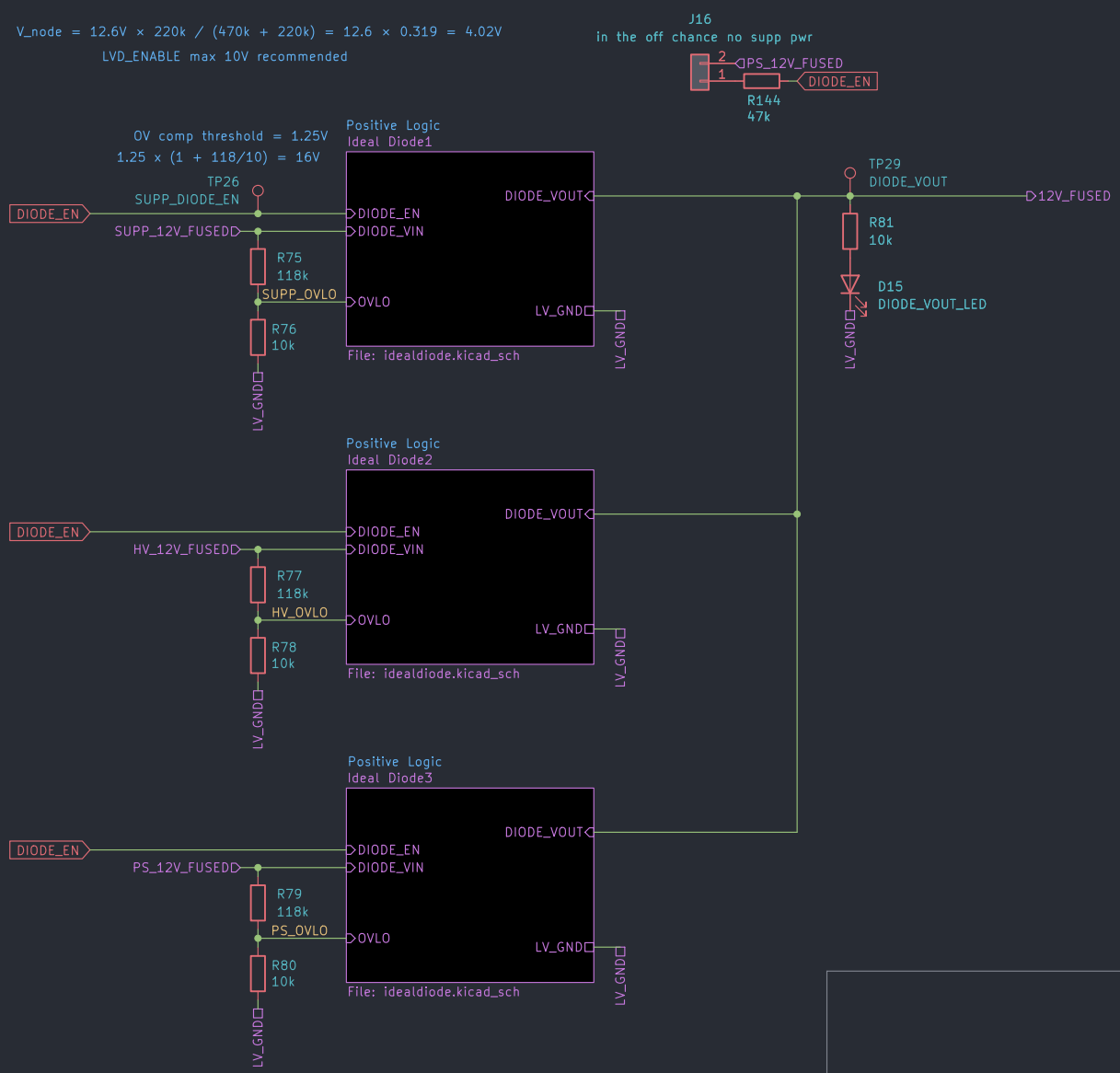
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 11/29

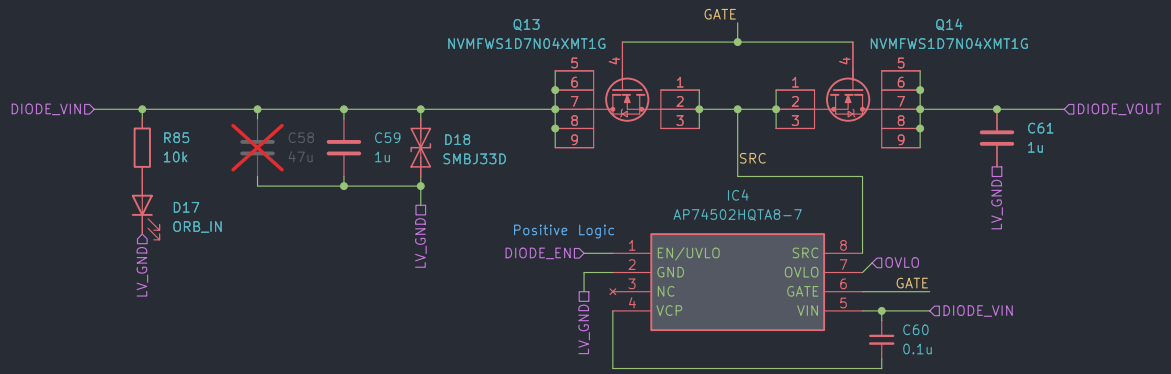


Sheet: /HV Input/HV -> 12V/Voltage + Current Sense/
File: sensing.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 12/29


V_node = 12.6V × 220k / (470k + 220k) = 12.6 × 0.319 = 4.02V
LVD_ENABLE max 10V recommended
J16
in the off chance no supp pwr
2
1
PS_12V_FUSED
DIODE_EN
R144
47k
OV comp threshold = 1.25V
1.25 x (1 + 118/10) = 16V
Positive Logic
Ideal Diode1
TP26
SUPP_DIODE_EN
DIODE_EN
SUPP_12V_FUSEDD
R75
118k
SUPP_OVLO
R76
10k
LV_GNDD
DIODE_EN
DIODE_VIN
OVLO
DIODE_VOUT
LV_GND
File: idealdiode.kicad_sch
LV_GNDD
TP29
DIODE_VOUT
12V_FUSED
R81
10k
D15
DIODE_VOUT_LED
LV_GNDD
Positive Logic
Ideal Diode2
DIODE_EN
HV_12V_FUSEDD
R77
118k
HV_OVLO
R78
10k
LV_GNDD
DIODE_EN
DIODE_VIN
OVLO
DIODE_VOUT
LV_GND
File: idealdiode.kicad_sch
LV_GNDD
Positive Logic
Ideal Diode3
DIODE_EN
PS_12V_FUSEDD
R79
118k
PS_OVLO
R80
10k
LV_GNDD
DIODE_EN
DIODE_VIN
OVLO
DIODE_VOUT
LV_GND
File: idealdiode.kicad_sch
LV_GNDD
Sheet: /Main Diode ORs/
File: diodeors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 13/29



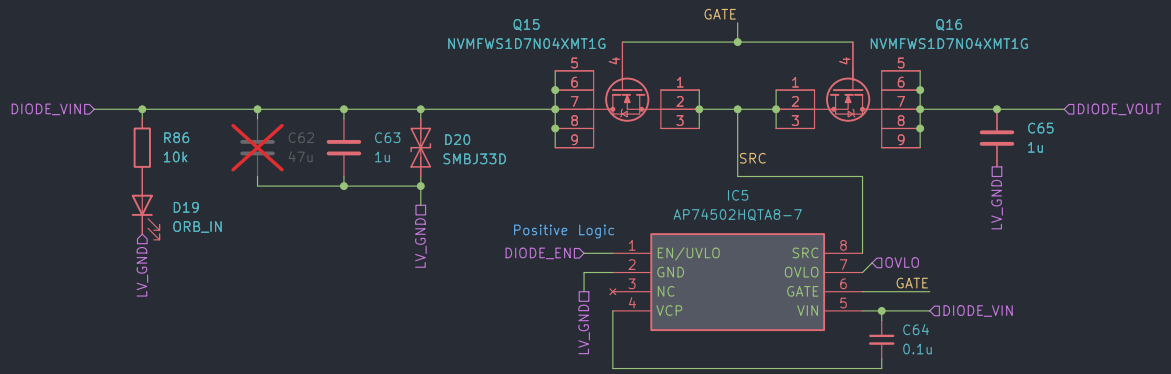
TP30
DIODE_VIN

DIODE_VIN

Sheet: /Main Diode ORs/Ideal Diode1/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 14/29



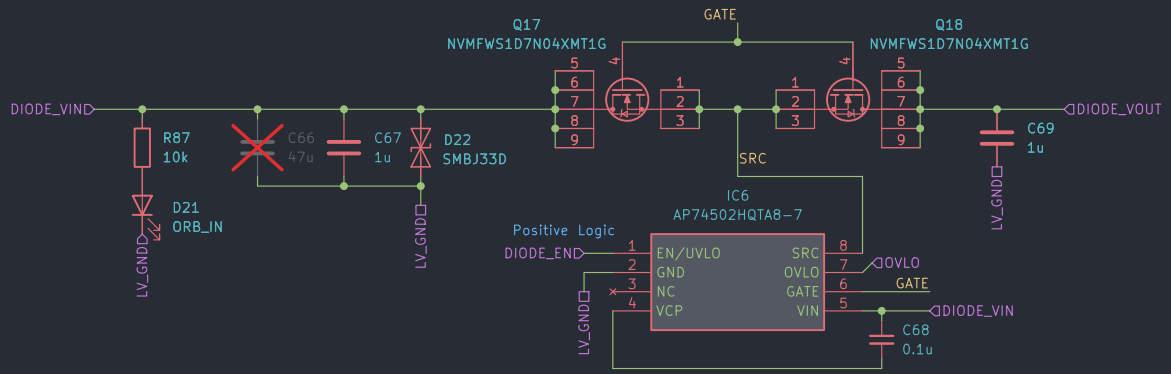
TP31
DIODE_VIN

DIODE_VIN

Sheet: /Main Diode ORs/Ideal Diode2/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 15/29



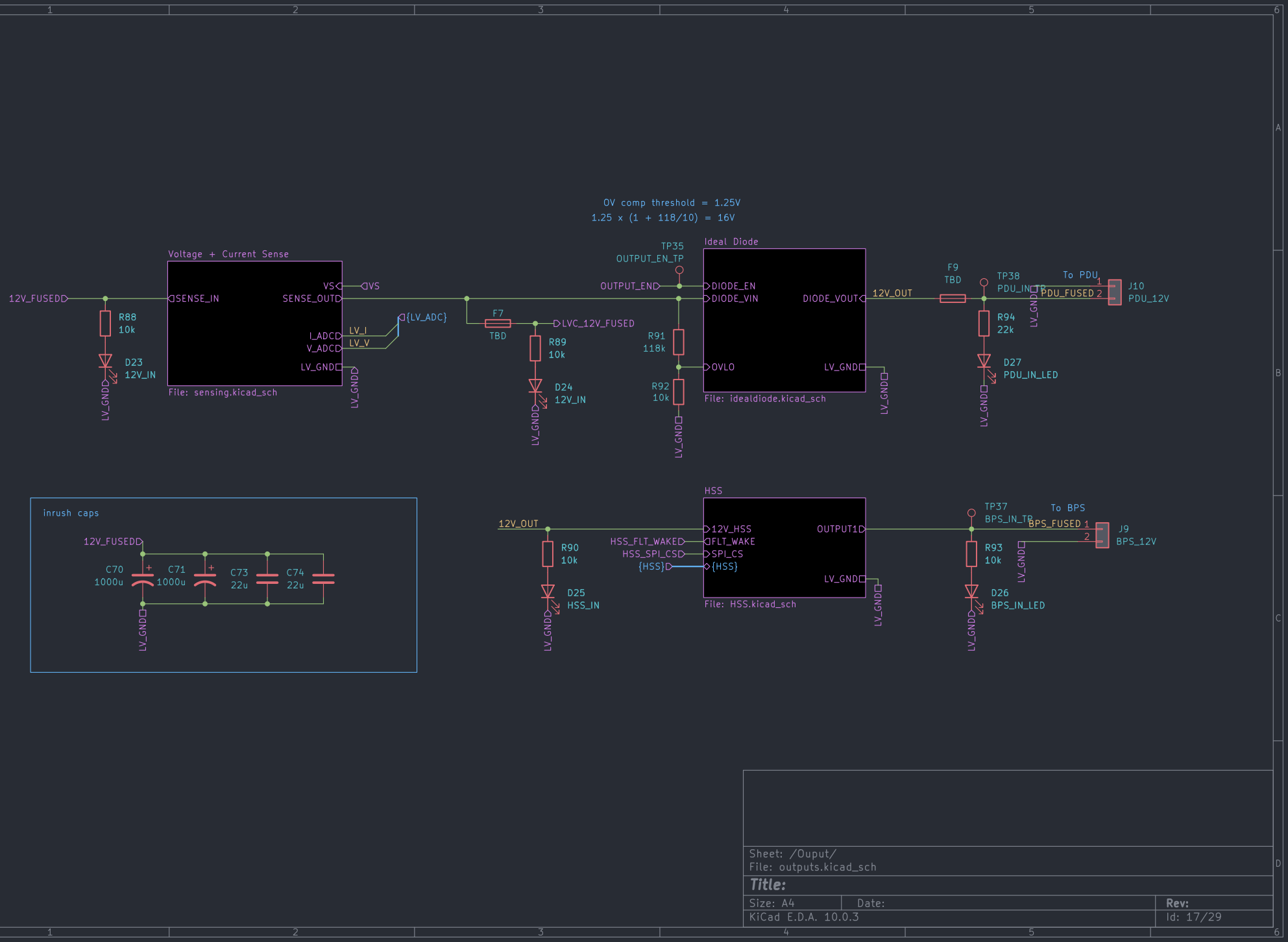
TP32
DIODE_VIN

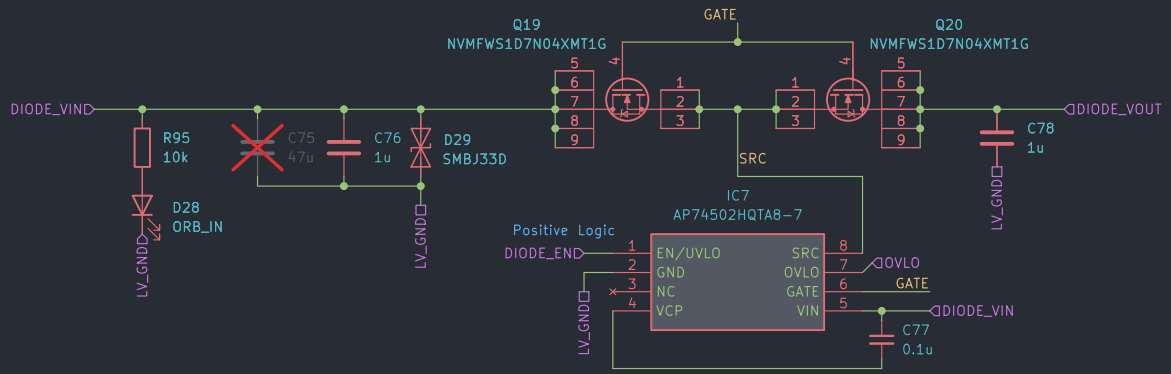
DIODE_VIN

Sheet: /Main Diode ORs/Ideal Diode3/
File: idealdiode.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 16/29


OV comp threshold = 1.25V
1.25 x (1 + 118/10) = 16V
Voltage + Current Sense
12V_FUSED
SENSE_IN
SENSE_OUT
VS
I_ADCD
V_ADCD
LV_GND
File: sensing.kicad_sch
R88
10k
D23
12V_IN
LV_I
LV_V
{LV_ADC}
F7
TBD
LVC_12V_FUSED
R89
10k
D24
12V_IN
TP35
OUTPUT_EN_TP
OUTPUT_EN
Ideal Diode
DIODE_EN
DIODE_VIN
DIODE_VOUT
OVLO
LV_GND
File: idealdiode.kicad_sch
R91
118k
R92
10k
12V_OUT
F9
TBD
TP38
PDU_IN_TP
To PDU
PDU_FUSED
J10
PDU_12V
R94
22k
D27
PDU_IN_LED
inrush caps
12V_FUSED
C70
1000u
C71
1000u
C73
22u
C74
22u
LV_GND
HSS
12V_OUT
12V_HSS
FLT_WAKE
SPI_CS
{HSS}
OUTPUT1
HSS_FLT_WAKE
HSS_SPI_CS
File: HSS.kicad_sch
R90
10k
D25
HSS_IN
TP37
BPS_IN_TP
To BPS
BPS_FUSED
J9
BPS_12V
R93
10k
D26
BPS_IN_LED
Sheet: /Ouput/
File: outputs.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 17/29



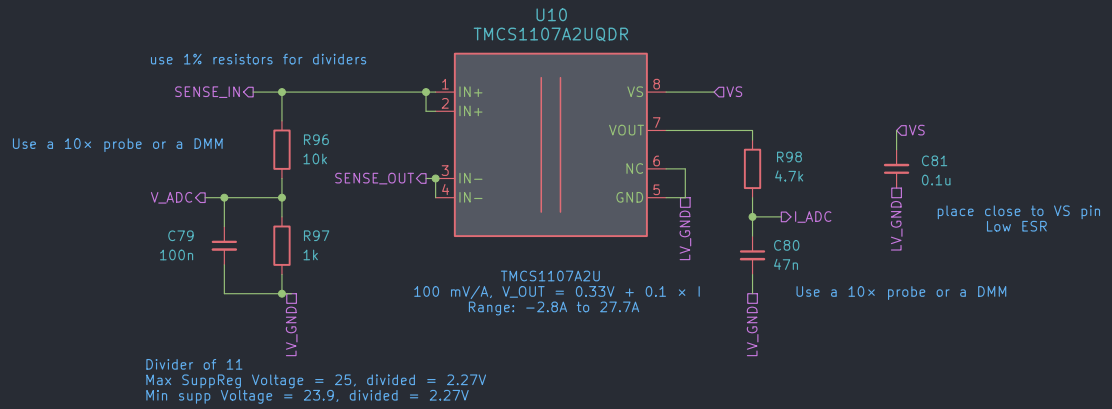
TP39
DIODE_VIN

DIODE_VIN

Sheet: /Ouput/Ideal Diode/
File: idealdiode.kicad_sch

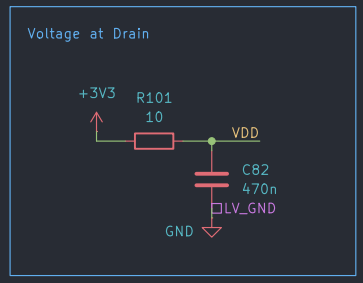
Title:

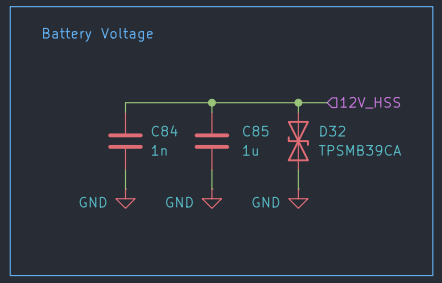
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KiCad E.D.A. 10.0.3		Id: 18/29

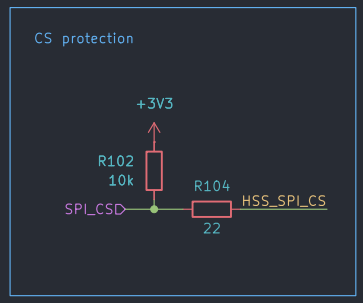


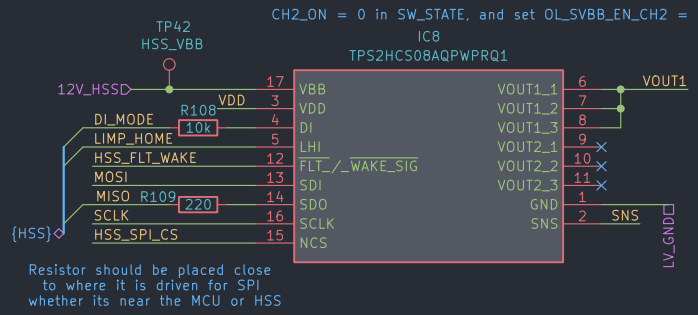
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File: sensing.kicad_sch		
Title:		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 19/29

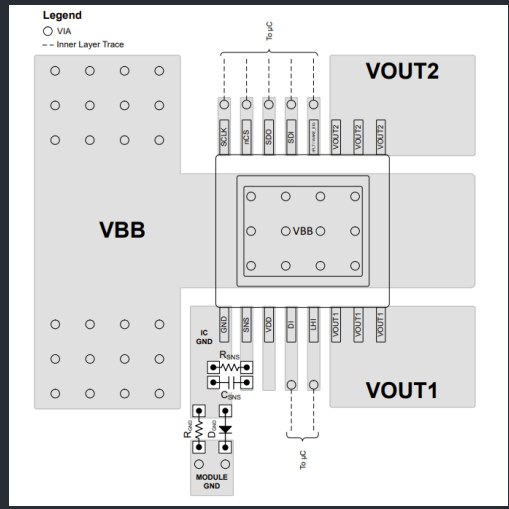
Pin Number	Pin Name	Type	Description
1	GND	GND	Device ground
2	SNS	O	SNS current output – use a parallel RC network to the GND pin of the IC.
3	VDD	Power	Logic Supply Input – closely decouple to GND pin of the IC with a ceramic capacitor.
4	DI	I	Sets the output behavior in the LIMP HOME mode, if configured as such.The pin needs to be connected to MCU or other HI/LO source through a 10K resistor tor protection and enabling the reverse polarity FET turn-on function.
5	LHI	I	External input (active High) to enable LIMP HOME mode.
6,7,8	VOUT1	O	Output of channel 1
9,10,11	VOUT2	O	Output of channel 2
12	$\overline{\text{FLT}}$ / $\overline{\text{WAKE_SIG}}$	O	Fault output – on any (one or more) channel - open drain, pull up with a 4.7K resistor to VDD pin. Also functions as a wake signal to the MCU upon load current demand in key-off mode.
13	SDI	I	SPI device (secondary) data input
14	SDO	O	SPI Data Output from the device. Internally pulled up to VDD.
15	CSN	I	Chip select. Internally pulled up to VDD
16	SCLK	I	SPI Clock input
Thermal Pad	VBB	Power	Power Supply

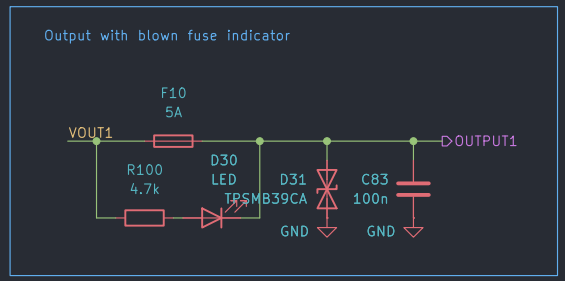


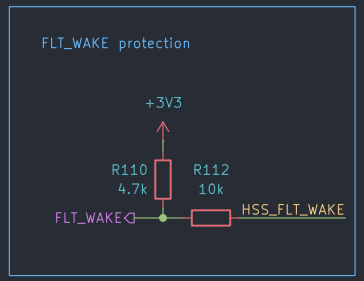


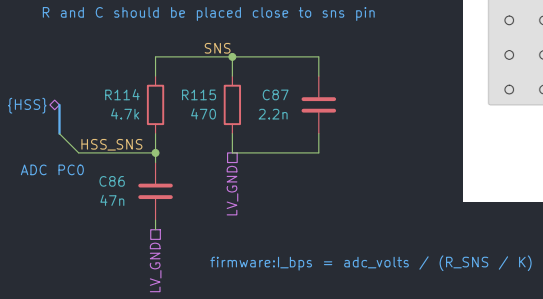


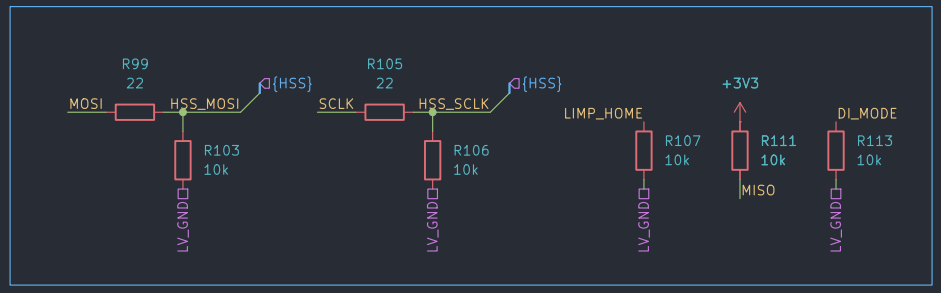










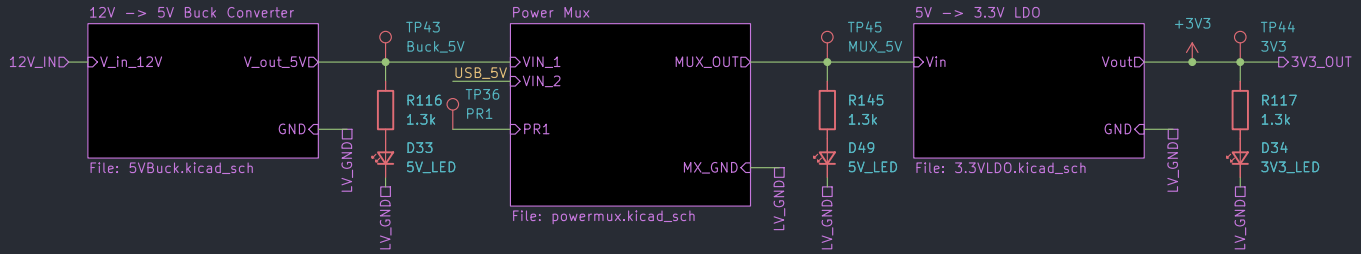


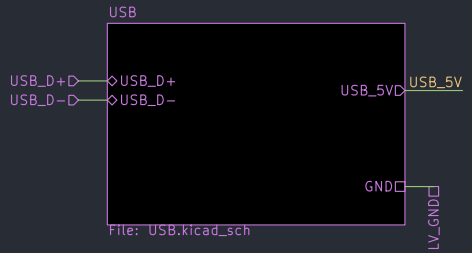
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Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 20/29



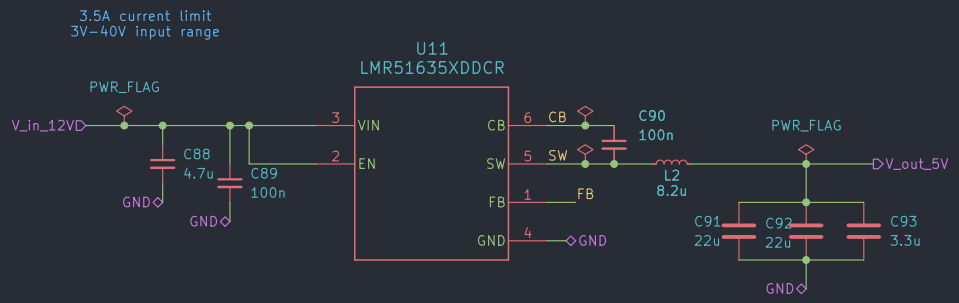


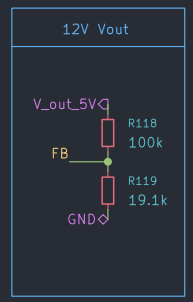
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File: auxpower.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 21/29

12V–5V Buck Converter



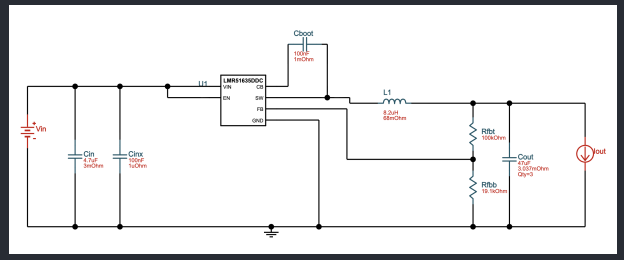


Fsw 400khz Vout 5

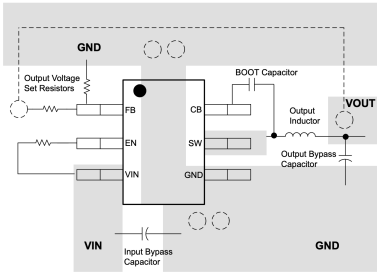
Table 8-2. Typical External Component Values for LMR51635, 3.5A Output Current

f_{SW} (kHz)	V_{OUT} (V)	L (µH)	C_{OUT} (1)	R_{FBT} (kΩ)	R_{FBB} (kΩ)	C_{FF}
400	3.3	5.6	2 × 47uF / 10V	31.6	10.2	Open
	5	10	2 × 33uF / 16V	53.6	10.2	Open
	12	15	10uF + 22uF / 35V	143	10.2	Open
1100	3.3	2.2	2 × 22uF / 10V	31.6	10.2	Open
	5	3.3	33uF / 16V	53.6	10.2	Open

Test Points



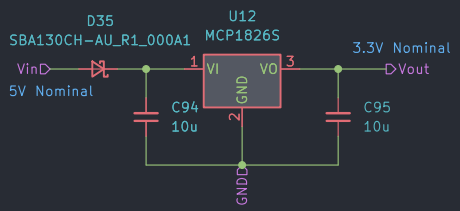
8.4.2 Layout Example



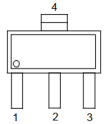
Sheet: /Power + Data/12V –> 5V Buck Converter/
File: 5VBuck.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 22/29



SOT-223-3



Pin	
1	V_{IN}
2	GND (TAB)
3	V_{OUT}
4	GND (TAB)

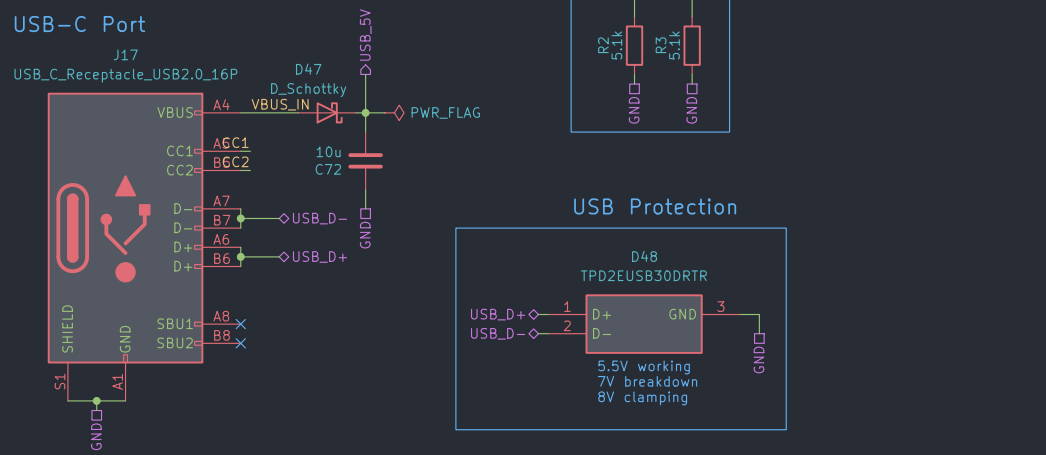
Parameters	Min	Nominal	Max	Unit	
Input Operating Voltage	2.3		6.0	V	
Input Quiescent Current		120	220	uA	
Max Ouput Current	1000			mA	
Line Regulation		+-0.05	+-0.20	%/V	
Load Regulation	-1.0	+-0.5	1.0	%	
Voltage Regulation	VR-2.5%	VR+-0.5%	VR+2.5%	V	VR=regulated output voltage=3.3V
Output Short Circuit Current		2.2		A	
Dropout Voltage		250	400	mV	

Input Nominal: 5V
Diode Forward Voltage: 500-600 mV
LDO Dropout Voltage Max: 400 mV
-> ~0.7V headroom on input

Sheet: /Power + Data/5V -> 3.3V LDO/
File: 3.3VLDO.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 23/29



Sheet: /Power + Data/USB/
File: USB.kicad_sch

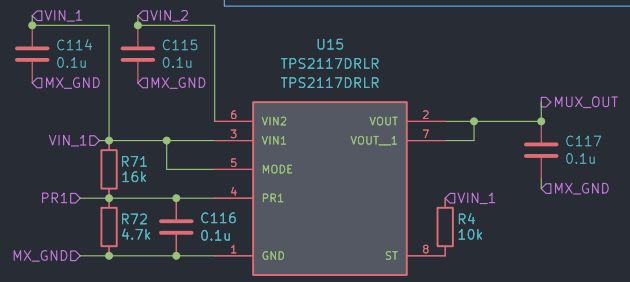
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 28/29

5V/3V3 Power Mux

Configured to use automatic switching mode: VIN_1 is prioritized

Acceptable input range is 1.6 to 5.5 volts
Resistor divider needs to be tuned depending on input



Resistor divider to meet Vref
User needs to set resistors!
Example values:
For 5v: R1 = 16k R2 = 4.7k (Max threshold = 4.76, min = 4.05)
For 3v3: R1 = 18k R2 = 10k (Max threshold = 3.02, min = 2.58)

Max and min threshold voltage calculations for 3v3 and 5v
[delete this if you want :^)]

3v3

$R_1 = 18000$ (−10 … 18000)

$R_2 = 10000$ (−10 … 10000)

$M = 1.08 \frac{(R_2 + R_1)}{R_2}$ = 3.024

$m = 0.92 \frac{(R_2 + R_1)}{R_2}$ = 2.576

5v

$R_1 = 16000$ (−10 … 16000)

$R_2 = 4700$ (−10 … 4700)

$M = 1.08 \frac{(R_2 + R_1)}{R_2}$ = 4.75659574468

$m = 0.92 \frac{(R_2 + R_1)}{R_2}$ = 4.05191489362

Sheet: /Power + Data/Power Mux/
File: powermux.kicad_sch

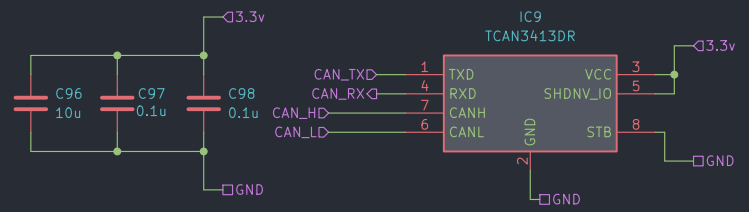
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 29/29

CAN Transceiver

7.3.1.5 V_{IO} (TCAN3413 only)

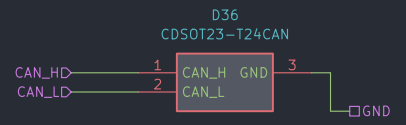
The V_{IO} pin provides the digital I/O voltage to match the CAN controller voltage thus avoiding the requirement for a level shifter. It supports voltages from 1.7 V to 3.6 V providing a wide range of controller support.



7.3.1.7 STB (Standby)

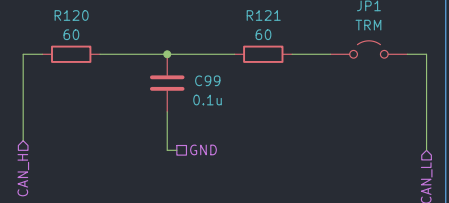
The STB pin is an input pin used for mode control of the transceiver. The STB pin can be supplied from either the system processor or from a static system voltage source. STB pin has default pull-up resistor on-chip. So if the pin is left floating or pulled high externally, device is in standby mode. If normal mode is the only intended mode of operation, the STB pin can be tied directly to GND.

CAN Protection



Try to keep CDSOT23 near bus conn

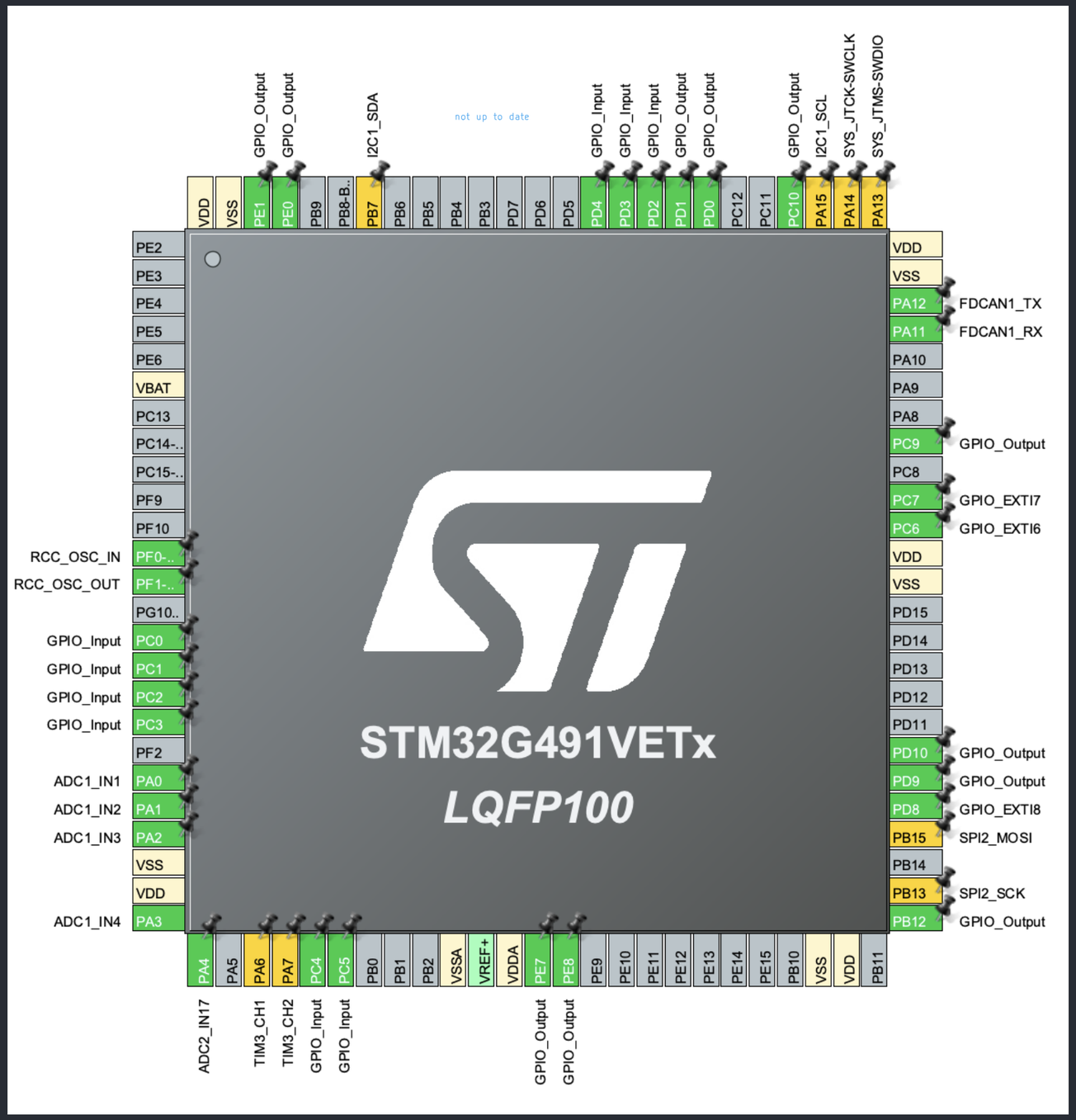
CAN Bus Termination

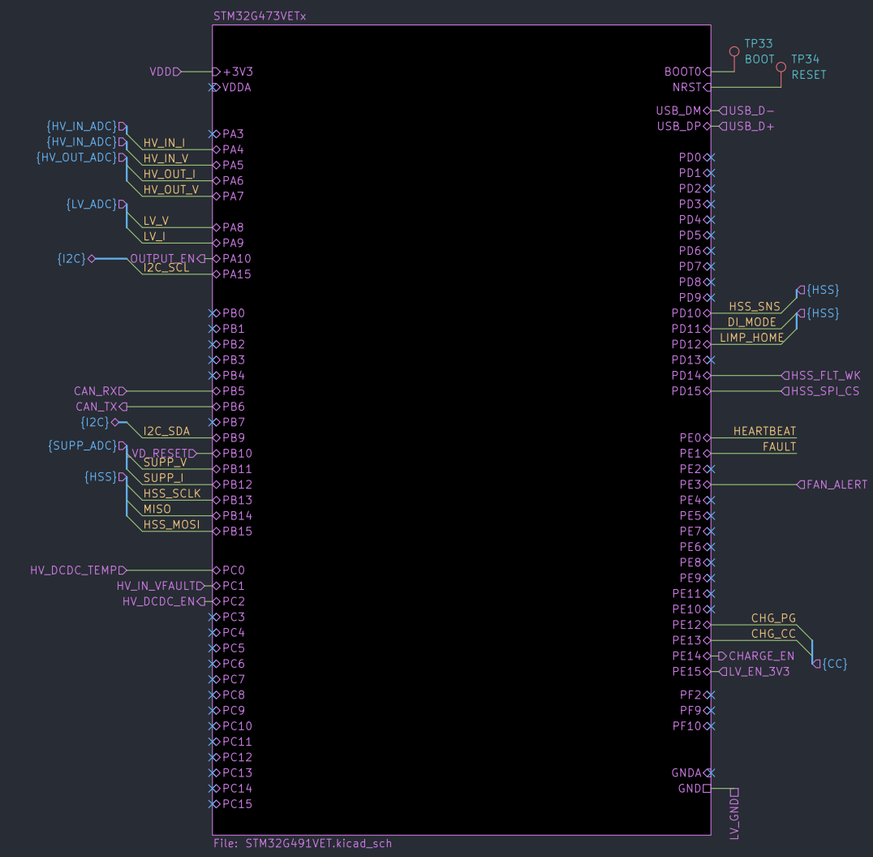


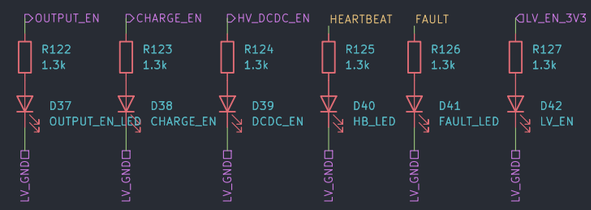
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Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 24/29







Sheet: /MCU/
File: MCU.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 25/29

100 Pin MCU

Use PA0, PA1, PA2 to PWM RGB LED
(TIM2_CH1, TIM2_CH2, TIM2_CH3)

BOOT0 low -> Load into app
BOOT0 high -> Load into bootloader

26p C100
26p C101
Y1 Crystal
*Place oscillator circuitry close to MCU

U13
STM32G473VETx

VDDA
+3V3
VREF+ 36
VDDA 37
VBAT 6
VDD 24
VDD 49
VDD 64
VDD 75
VDD 100

NRST 14
BOOT0 95

PG10-NRST
PB8-BOOT0
PF0-OSC_IN 12
PF1-OSC_OUT 13
PF2 19
PF9 10
PF10 11

PA0-GPIO_BLUE 20 → PA0 BLUE
PA1-GPIO_RED 21 → PA1 RED
PA2-GPIO_GREEN 22 → PA2 GREEN
PA3 25
PA4 26
PA5 27
PA6 28
PA7 29
PA8 69
PA9 70
PA10 71
PA11 72 USB_DM
PA12 73 USB_DP
PA13 76 SWDIO
PA14 77 SWCLK
PA15 78

PB0 32
PB1 33
PB2 34
PB3 90
PB4 91
PB5 92
PB6 93
PB7 94
PB9 96
PB10 47
PB11 50
PB12 51
PB13 52
PB14 53
PB15 54

PC0 15
PC1 16
PC2 17
PC3 18
PC4 30
PC5 31
PC6 65
PC7 66
PC8 67
PC9 68
PC10 79
PC11 80
PC12 81
PC13 7
PC14-OSC32_IN 8
PC15-OSC32_OUT 9

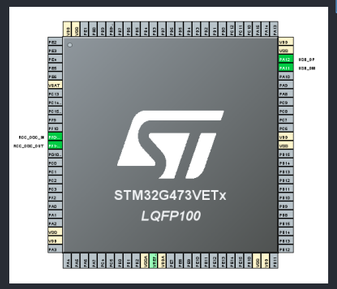
PE0 97
PE1 98
PE2 1
PE3 2
PE4 3
PE5 4
PE6 5
PE7 38
PE8 39
PE9 40
PE10 41
PE11 42
PE12 43
PE13 44
PE14 45
PE15 46

PD0 82
PD1 83
PD2 84
PD3 85
PD4 86
PD5 87
PD6 88
PD7 89
PD8 55
PD9 56
PD10 57
PD11 58
PD12 59
PD13 60
PD14 61
PD15 62

VSSA 35
VSS 23
VSS 48
VSS 63
VSS 74
VSS 99
GNDA
FB1 FerriteBead
GND

Beware, using PC13 – 15 as GPIOs ...

2. PC13, PC14 and PC15 are supplied through the power switch. Since the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is limited:
   - The speed should not exceed 2 MHz with a maximum load of 30 pF
   - These GPIOs must not be used as current sources (e.g. to drive an LED).
3. After a backup domain power-up, PC13, PC14 and PC15 operate as GPIOs. Their function then depends on the content of the RTC registers which are not reset by the system reset. For details on how to manage these GPIOs, refer to the Backup domain and RTC register descriptions in the reference manual RM0440 "STM32G4 Series advanced Arm®-based 32-bit MCUs".



Power Source Filtering/Isolation

+3V3
VDD Bypass Caps
C102 10u
C104 0.1u
C107 0.1u
C109 0.1u
C111 0.1u
C112 0.1u
GNDD
Place as close as possible to MCU VDD pins.

VDDA
VDDA Bypass Caps
C103 0.1u
C105 0.1u
C108 1u
C110 1u
GNDA
Place near VDDA/VSSA pins.

VDDA Separation
FB2 FerriteBead
+3V3
VDDA
C113 0.1u
GNDA

Place these decoupling caps as close as possible to their respective leads.

Reset BTN + Boot SW

+3V3
R128 10k
NRST
C106 0.1u
Reset1 Reset
GNDD
Reset Button

Sys Memory (bootloader)
SW_SPDT SW1
+3V3
GND
Main Flash (user code)
D43 PESD18VL1BA
R134 10k
D45 LED
GND
BOOT0
R135 1k
GND

Serial Wire Communication

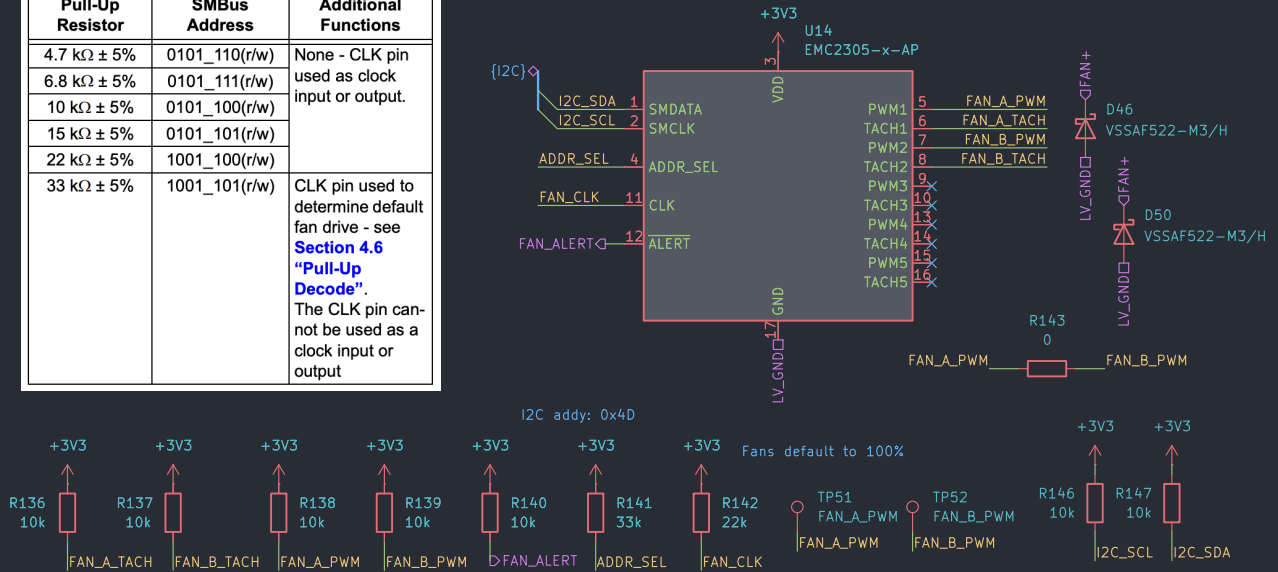
Conn J12 SWD_Header
+3V3D
GND
3V3
GND
SWCLK
SWDIO
R129 22
SWCLK
SWDIO
R130 22
J13 SWD
+3V3D
GND
CLK
IO
Pin Headers

RGB LED

PA2 PA1 PA0
B G R
D44 RGB_LED
R131 1k
R132 1k
R133 1k
GND

Sheet: /MCU/STM32G473VETx/
File: STM32G491VET.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.3 | Id: 26/29

Pull-Up Resistor	SMBus Address	Additional Functions
4.7 kΩ ± 5%	0101_110(r/w)	None - CLK pin used as clock input or output.
6.8 kΩ ± 5%	0101_111(r/w)	
10 kΩ ± 5%	0101_100(r/w)	
15 kΩ ± 5%	0101_101(r/w)	
22 kΩ ± 5%	1001_100(r/w)	
33 kΩ ± 5%	1001_101(r/w)	CLK pin used to determine default fan drive - see **Section 4.6 "Pull-Up Decode"**. The CLK pin cannot be used as a clock input or output



Sheet: /Fans/
File: tempcontrol.kicad_sch

Title:

Size: A5 Date:

KiCad E.D.A. 10.0.3

TABLE 4-9: CLK PIN PULL-UP DECODE

Pull-Up Resistor	Fan Default Drive Setting
4.7k Ohm ±5%	0% - OFF
6.8k Ohm ±5%	30%
10k Ohm ±5%	50%
15k Ohm ±5%	75%
22k Ohm ±5%	100%
33k Ohm ±5%	0% - OFF